

ASSIGNMENT NO.4. Working with Multiple Tables and Joins

Questions:

Q1) Create a MySQL database named CollegeDB and create the following tables:

Student Table:

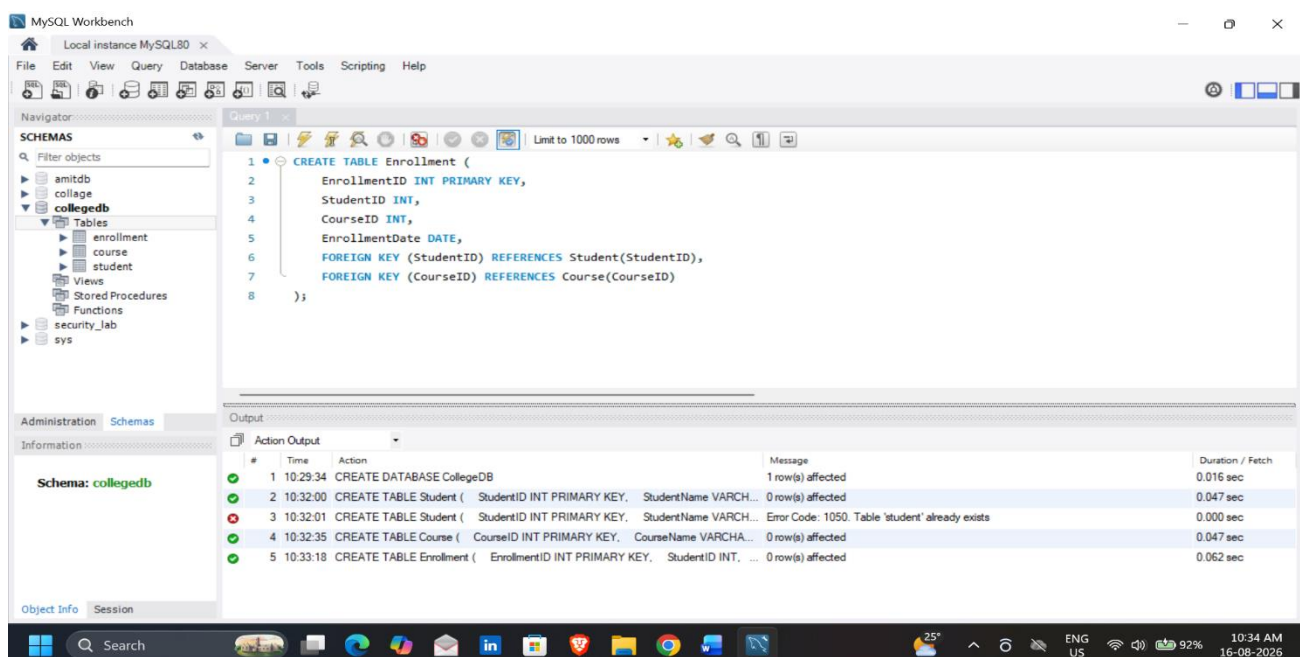
Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
City	VARCHAR(30)	

Course Table:

Column Name	Data type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	
Fees	DECIMAL(10,2)	

Enrollment Table:

Column Name	Data type	Constraint
EnrollmentID	INT	PRIMARY KEY
StudentID	INT	FOREIGN KEY
CourseID	INT	FOREIGN KEY
EnrollmentDate	DATE	



Q2) Insert at least 6 records into the student table, 4 records into the Course table, and 5 records into the Enrollment table.

Make sure:

Requirement

At least one student should not have enrollment

At least one course should not have any enrolled student

MySQL Workbench interface showing the 'collegedb' schema. The 'student' table is selected, and the 'Result Grid' displays 6 records:

StudentID	StudentName	City
1	Rahul Sharma	Mumbai
2	Priya Patil	Pune
3	Aman Verma	Nashik
4	Sneha Joshi	Mumbai
5	Karan Mehta	Thane
6	Neha Shah	Pune

The 'Output' pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
4	10:32:35	CREATE TABLE Course (CourseID INT PRIMARY KEY, CourseName VARCHAR(50))	0 row(s) affected	0.047 sec
5	10:33:18	CREATE TABLE Enrollment (EnrollmentID INT PRIMARY KEY, StudentID INT, CourseID INT)	0 row(s) affected	0.062 sec
6	10:51:17	INSERT INTO Student (StudentID, StudentName, City) VALUES (1, 'Rahul Sharma', 'Mumbai')	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.016 sec
7	10:51:44	SELECT * FROM Student LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench interface showing the 'collegedb' schema. The 'course' table is selected, and the 'Result Grid' displays 4 records:

CourseID	CourseName	Fees
101	BSc IT	45000.00
102	BCA	40000.00
103	BSc CS	42000.00
104	Data Science	55000.00

The 'Output' pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
6	10:51:17	INSERT INTO Student (StudentID, StudentName, City) VALUES (1, 'Rahul Sharma', 'Mumbai')	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.016 sec
7	10:51:44	SELECT * FROM Student LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
8	10:52:54	INSERT INTO Course (CourseID, CourseName, Fees) VALUES (101, 'BSc IT', 45000.00)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
9	10:53:10	SELECT * FROM Course LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x

Limit to 1000 rows

1 • `SELECT * FROM Enrollment;`

Result Grid

EnrollmentID	StudentID	CourseID	EnrollmentDate
1	1	101	2026-07-01
2	2	102	2026-07-02
3	3	103	2026-07-03
4	4	101	2026-07-04
5	5	102	2026-07-05
NULL	NULL	NULL	NULL

Administration Schemas

Information: Enrollment 3 x

Schema: collegedb

Output: Action Output

#	Time	Action	Message	Duration / Fetch
8	10:52:54	INSERT INTO Course (CourseID, CourseName, Fees) VALUES (101, 'BSc IT', 450...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
9	10:53:10	SELECT * FROM Course LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
10	10:54:05	INSERT INTO Enrollment (EnrollmentID, StudentID, CourseID, EnrollmentDate) VA...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	10:54:17	SELECT * FROM Enrollment LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

25° ENG US 98% 10:54 AM 16-08-2026

Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses.

The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x

Limit to 1000 rows

1 `SELECT`

2 `Student.StudentID,`

3 `Student.StudentName,`

4 `Enrollment.CourseID,`

5 `Enrollment.EnrollmentDate`

6 `FROM Student`

7 `INNER JOIN Enrollment`

8 `ON Student.StudentID = Enrollment.StudentID;`

Result Grid

StudentID	StudentName	CourseID	EnrollmentDate
1	Rahul Sharma	101	2026-07-01
2	Priya Patil	102	2026-07-02
3	Aman Verma	103	2026-07-03
4	Sneha Joshi	101	2026-07-04
5	Karan Mehta	102	2026-07-05

Administration Schemas

Information: Result 4 x

Schema: collegedb

Output: Action Output

#	Time	Action	Message	Duration / Fetch
11	10:54:17	SELECT * FROM Enrollment LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	11:00:05	SELECT Student.StudentID, Student.StudentName, Enrollment.CourseID, ...	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

25° ENG US 99% 11:00 AM 16-08-2026

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables.

Use:

Tables Required
Student
Course
Enrollment

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'collegedb' database selected. The 'Query Editor' in the center contains the following SQL query:

```

1  SELECT
2      Student.StudentName,
3      Student.City,
4      Course.CourseName,
5      Course.Fees,
6      Enrollment.EnrollmentDate
7  FROM Enrollment
8  INNER JOIN Student
9  ON Enrollment.StudentID = Student.StudentID
10 INNER JOIN Course
11 ON Enrollment.CourseID = Course.CourseID;
    
```

Below the query editor, the 'Result Grid' displays the results of the query. The results are as follows:

StudentName	City	CourseName	Fees	EnrollmentDate
Rahul Sharma	Mumbai	BSc IT	45000.00	2026-07-01
Sneha Joshi	Mumbai	BSc IT	45000.00	2026-07-04
Priya Patil	Pune	BCA	40000.00	2026-07-02
Karan Mehta	Thane	BCA	40000.00	2026-07-05
Aman Verma	Nashik	BSc CS	42000.00	2026-07-03

At the bottom, the 'Output' pane shows the execution details: 5 row(s) returned, 0.000 sec / 0.000 sec.

Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course.

The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```

1 SELECT
2     Student.StudentID,
3     Student.StudentName,
4     Student.City,
5     Enrollment.CourseID,
6     Enrollment.EnrollmentDate
7 FROM Student
8 LEFT JOIN Enrollment

```

The Result Grid displays the following data:

StudentID	StudentName	City	CourseID	EnrollmentDate
1	Rahul Sharma	Mumbai	101	2026-07-01
2	Priya Patil	Pune	102	2026-07-02
3	Aman Verma	Nashik	103	2026-07-03
4	Sneha Joshi	Mumbai	101	2026-07-04
5	Karan Mehta	Thane	102	2026-07-05
6	Neha Shah	Pune	101	2026-07-05

The Action Output window shows the following details:

#	Time	Action	Message	Duration / Fetch
13	11:06:12	SELECT	Student.StudentName, Student.City, Course.CourseName, Cour...	5 row(s) returned 0.000 sec / 0.000 sec
14	11:12:59	SELECT	Student.StudentID, Student.StudentName, Student.City, Enrollm...	6 row(s) returned 0.000 sec / 0.000 sec

Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students.

The output should include:

Column
StudentName
CourseID
CourseName
Fees

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```

1 SELECT
2     Student.StudentName,
3     Course.CourseID,
4     Course.CourseName,
5     Course.Fees
6 FROM Enrollment
7 RIGHT JOIN Course
8 ON Enrollment.CourseID = Course.CourseID
9 LEFT JOIN Student
10 ON Enrollment.StudentID = Student.StudentID;

```

The Result Grid displays the following data:

StudentName	CourseID	CourseName	Fees
Rahul Sharma	101	BSc IT	45000.00
Sneha Joshi	101	BSc IT	45000.00
Priya Patil	102	BCA	40000.00
Karan Mehta	102	BCA	40000.00
Aman Verma	103	BSc CS	42000.00
	104	Data Science	55000.00

The Action Output window shows the following details:

#	Time	Action	Message	Duration / Fetch
14	11:12:59	SELECT	Student.StudentID, Student.StudentName, Student.City, Enrollm...	6 row(s) returned 0.000 sec / 0.000 sec